ModelIt!

Student Activity Pack

Why Does a Black Car Get Hotter?

How Surface Color Affects Heat Absorption from Sunlight

Grade 04L2 | NGSS 4-PS3-2 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Have you ever touched a dark car seat on a hot day? What did it feel like compared to a lighter colored seat?**

2. **Why do you think people in very hot countries often wear white or light-colored clothing?**

3. **Draw what you think happens when sunlight hits a black surface versus a white surface.**

4. **Do you think a dark playground slide or a light playground slide gets hotter in the sun? Why?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Sunlight Intensity | Surface Color | Heat Absorption | Surface Temperature | Comfort Level

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Bright Sun, Dark Surface

Sunlight Intensity at 80%, Surface Color at dark (80%)

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Bright Sun, Light Surface

Sunlight Intensity at 80%, Surface Color at light (20%)

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Cloudy Day, Dark Surface

Sunlight Intensity at 20%, Surface Color at dark (80%)

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Wind Speed — How fast the air moves across the surface. Wind carries heat away from surfaces, cooling them down even if they absorb a lot of light energy

Surface Material — What the surface is made of. Metal heats up faster than wood or plastic, even if they are the same color

Shade Coverage — How much of the surface is blocked from direct sunlight by trees, buildings, or covers. Shade dramatically reduces heat absorption

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If two cars sit in the same sunny parking lot, why does the black car feel burning hot while the white car stays cooler? Use evidence from your simulation to support your answers.

1. **Why does a black car get hotter than a white car sitting in the same parking lot?**

2. **If energy is being absorbed by the dark surface, where does that energy originally come from?**

3. **Why do you think rooftops in hot cities are sometimes painted white?**

4. **If you were designing a playground for a very sunny area, what colors would you choose and why?**

5. **How did your model help you understand the connection between color and temperature?**

8 STEM Challenge: Design the Coolest Parking Lot

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YOUR ENGINEERING MISSION

Design a model parking lot surface that keeps cars and people as cool as possible on a hot sunny day.

THE SCENARIO

A school in a very hot city has a problem: their asphalt parking lot gets so hot in the summer that students cannot walk across it comfortably, car interiors become dangerously hot, and the playground equipment burns children's hands. The principal has hired your engineering team to redesign the parking lot surface to reduce temperatures. Use evidence from your model to recommend colors and materials.

Engineering Design Thinking

1. WHAT MATERIALS WILL YOU TEST (BLACK PAPER, WHITE PAPER, ALUMINUM FOIL, COLORED FABRIC)?

2. HOW WILL YOU MEASURE THE TEMPERATURE OF EACH SURFACE AFTER SITTING IN THE SUN?

3. HOW LONG WILL YOU LEAVE EACH SURFACE IN THE SUN BEFORE MEASURING?

4. HOW WILL YOU MAKE SURE ALL SURFACES GET THE SAME AMOUNT OF SUNLIGHT FOR A FAIR TEST?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Environmental Engineers design buildings, roads, and cities to stay cool and save energy. They choose surface colors and materials that reduce heat absorption, helping fight the 'urban heat island' effect where cities get hotter than surrounding areas. They earn $65,000-$110,000/year.